

2012

HiggsQCDHiggsHiggsPDFs; jetCMSATLASHiggs [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]

CMS13 TeVCMS13 TeV137.1 fb¹125.09 GeV13 TeV2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(stat.)^{+0.24}_{-0.19}(syst.) fb.
TeVQCD

SM)??0Higgs0HiggsHiggsHiggs [14, 15, 16, 17, 18, 19] SU(3)_C⊗SU(2)_L⊗U(1)_γ 1961Glashow
SU(2)_L⊗U(1)_γ Brout, Englert Higgs Higgs SalamWeinbergGlashow SU(2)_L⊗U(1)_γLY
?? GlashowHiggsWZ [14, 15, 16, 17, 18, 19]Higgs??Higgs2012LHCCMSATLAS

LHC-

LHCuudHiggsLHC- ??underlying-MPI) jet?? Factorization (μ_F) Factorization
??

5HiggsLHCHiggs-HiggstopVBFHiggsWZHiggsVBFjetW/ZHiggsWZWZLEPHiggsTevatronLE
topttHiggs

125 GeV Higgs1.6 × 10⁻²² sHiggs ?? [20]H → b**̄**m_H ≈ 125 GeVQCDCMSAT-
LAS [21], [22] [23]CDFD0BjetjetHiggsHiggs125GeVHiggsWWH → ZZ → ℓℓℓ
[20]HiggsH → ZZ → ℓℓℓ

TevatronLEPHiggs2010LHCCERN-7TeVℒ = 10³⁴cm²s⁻¹10010000LHC1998200875-
1503.8271983-1988LEPLHC

2012LHCCMSATLASHiggs [24, 25]HiggsHiggsHiggs m_H = 125.09±0.21(stat.)±
0.11(syst.) GeV [26]

20132014LHC2015LHC13TeV14TeV2015CMS13TeV4.2fb⁻¹201620172018CMS41.0
fb⁻¹, 49.8 fb⁻¹, 67.9 fb⁻¹LHCLHCHL-LHC3)Linac4Linac29010H-)160 MeVLHC-
CMSATLASALICELHCb ATLASpileupLAr CMSHPDSiPM) HL-LHCCSCFE ALI-
CEITSTPSDAQHLT)MFT LHCb1 MHz40 MHzFE

CMSCMSCMS ??LHCCMSCMS1 TeVCMSbMET100GeVTeV [27]CMS28 m7 m15
m14000

CMSITSECALHICALCMS3.0CMS

LHC400040 MHzDAQCMSL1HLT)

LHCggHVBFWHZH ttH)POWHEG V2WHZHPowheg MINLO HVJ [28]JHUGEN[29]WHZH
ZZ → 2ℓ2XWZPYTHIA8.209QCDPOWHEGNNPDF 3.1 NLO PDF [30] [12] ???ggHPowhegNNLO
V2PYTHIA8NLOq $\bar{q}$ → ZZ → 4ℓgg → ZZMCFMLO?? and ?? SIP_{3D} < 4 (SIP_{3D}isolation
cut)Z(e^+e^- , $\mu^+\mu^-$)12GeV120GeVZZZZZZ₁Z₂ZZ??
pp →H→ 4ℓ [31] [32] [33] [34]

13TeV2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(stat.)^{+0.24}_{-0.19}(syst.) fb [35]

POWHEGNNLOPS